Tristan Habémont

Graduate Engineer in ML, Computer Vision and Electrical Engineering

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EDUCATION

Feb 2025 - Mar 2026

Master 2 - MSc in Machine Learning and Medical Imaging
Centrale Lyon
- Ranked 2nd in cohort (GPA: 3.90/4)
- Completed alongside my final year at INSA Lyon

Sep 2022 - Mar 2026

Diplôme d’Ingénieur - MSc in Electrical Engineering
INSA Lyon
- Major in **Deep Learning, Image and Signal Processing** (GPA: 4.00/4)
- Ranked in the top 10% of cohort (GPA: 3.76/4)

Sep 2024 - Dec 2024

Exchange semester
Tōhoku University
- Completed the intensive Japanese language program
- Mentored bachelor students on Computer Vision projects

Sep 2020 - Jul 2022

DUT - University Diploma of Technology in Electrical Engineering
Université de Lorraine
- Major in **Multitasking and OOP** (GPA: 4.00/4)
- Ranked 1st in cohort, with Highest Honors (GPA: 3.89/4)

EXPERIENCE

Oct 2025 - Jan 2026
4 months

Research Project - Conditional Latent Diffusion for Dental Prediction
BovoPredict, in collab. with INSA Lyon - Supervised by Dr. Thomas Grenier and Dr. Chantal Muller
- Built a VAE + Latent Diffusion pipeline for dentate-to-edentulous CBCT slice generation
- Used self-supervised latent learning and diffusion denoising to guide anatomical reconstruction
- Designed Transformers-based multimodal conditioning for morphology metrics injection
- Developed t-SNE/UMAP latent-space analysis and morphology-based evaluation tools

Feb 2025 - Jul 2025
6 months

Research Intern - Graph ML & 3D Computer Vision
CREATIS, Lyon - Supervised by Dr. Odyssée Merveille and Prof. Olivier Bernard
- Trained and compared nnU-Net 3D segmentation models for cardiac ventricle analysis
- Researched and trained GNN architectures for pulmonary embolism risk stratification
- Built PyTorch Geometric vascular graph datasets from segmentation-derived 3D anatomy
- Co-authored a MICCAI 2026 submission, presented a poster and published a lab blog post

Feb 2024 - Jul 2024
6 months

Research Intern - Real-Time Deep Learning for Noisy Time-Series
Institut des Nanotechnologies de Lyon - Supervised by Dr. Bertrand Massot
- Researched robust real-time ECG/EDA peak detection methods for physiological VR monitoring
- Implemented ECG localization pipelines combining U-Net, LSTM and wavelet-based models
- Developed modular Python packages for BLE sensor acquisition and signal analytics
- Co-authored an IEEE VRW 2025 publication

Oct 2023 - Jan 2024
4 months

Technical Project - Embedded Computer Vision for Autonomous Drone Control
INSA Lyon - Supervised by Dr. Jean-François Mogniotte
- Designed a dual-ESP32 drone architecture with onboard CNN gesture recognition
- Developed real-time C/C++ firmware for sensor acquisition and SPI/I2C communication
- Implemented Kalman filtering for altitude estimation using ToF, barometer and IMU sensors
- Designed a custom drone PCB and monitoring dashboard for hardware debugging

Apr 2022 - Jul 2022
4 months

R&D Intern - Embedded Systems
Institut Jean Lamour, Nancy
- Developed STM32 serial communication modules in C/C++
- Controlled acoustic levitation device with configurable PWM
- Built Python API for device monitoring and wrote documentation

Sep 2021 - Apr 2022
8 months

Academic Tutor - Mathematics & Applied Physics
Université de Lorraine
- Teaching and support for first-year students
- Courses: complex analysis, electrostatics and electromagnetism

TECH KEYWORDS

AI/ML	PyTorch/Lightning, scikit-learn, pandas, TensorFlow/Keras, Hydra, W&B, OpenCV
EMBEDDED SYSTEMS	C/C++, VHDL, RTOS, PlatformIO, TFLite, STM32-HAL, ESP-IDF, RPi OS
SOFTWARE & TOOLS	Java, C#, SQL, Linux, Git, GitHub CI/CD, Bash, Docker, HPC/Slurm

LANGUAGES

French: Native | English: C1 | Japanese: N5